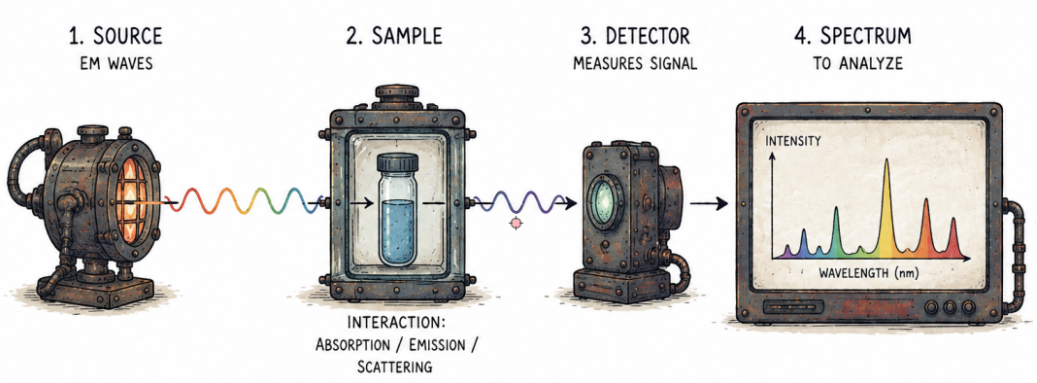


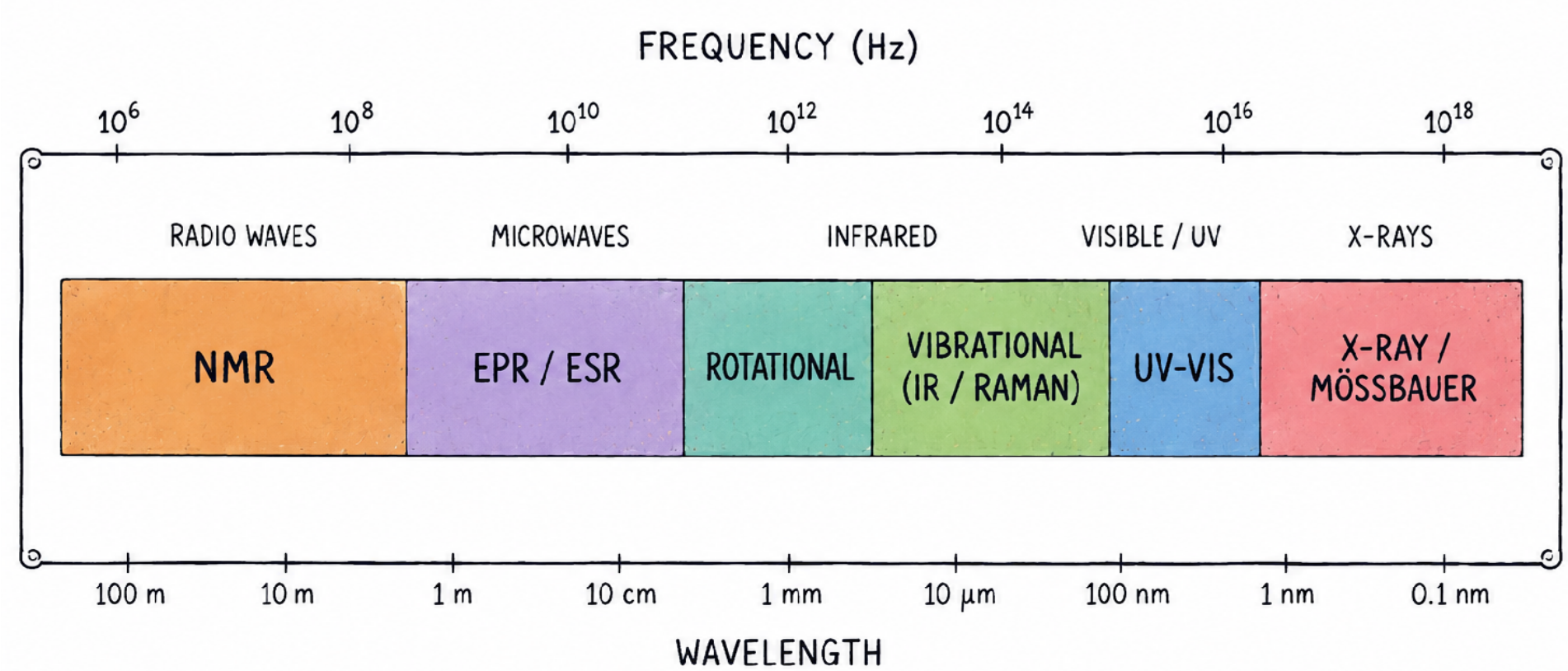
NMR Spectroscopy with Quantum computers

Andisheh Khedri

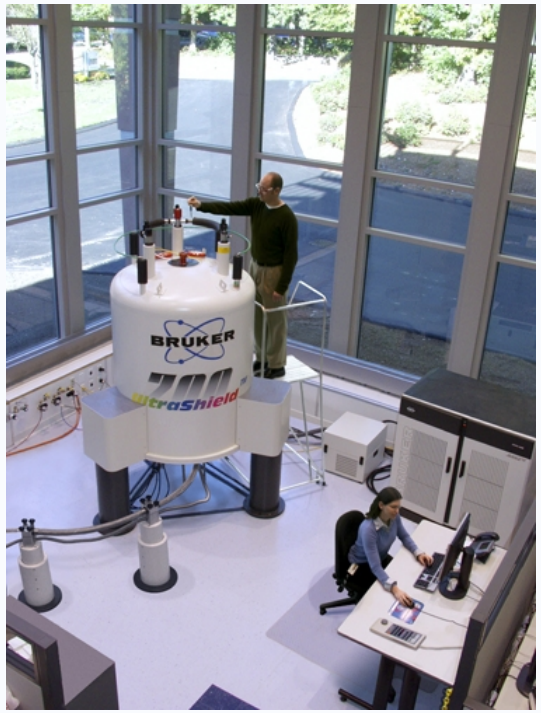
HQS QUANTUM SIMULATIONS

Spectroscopy

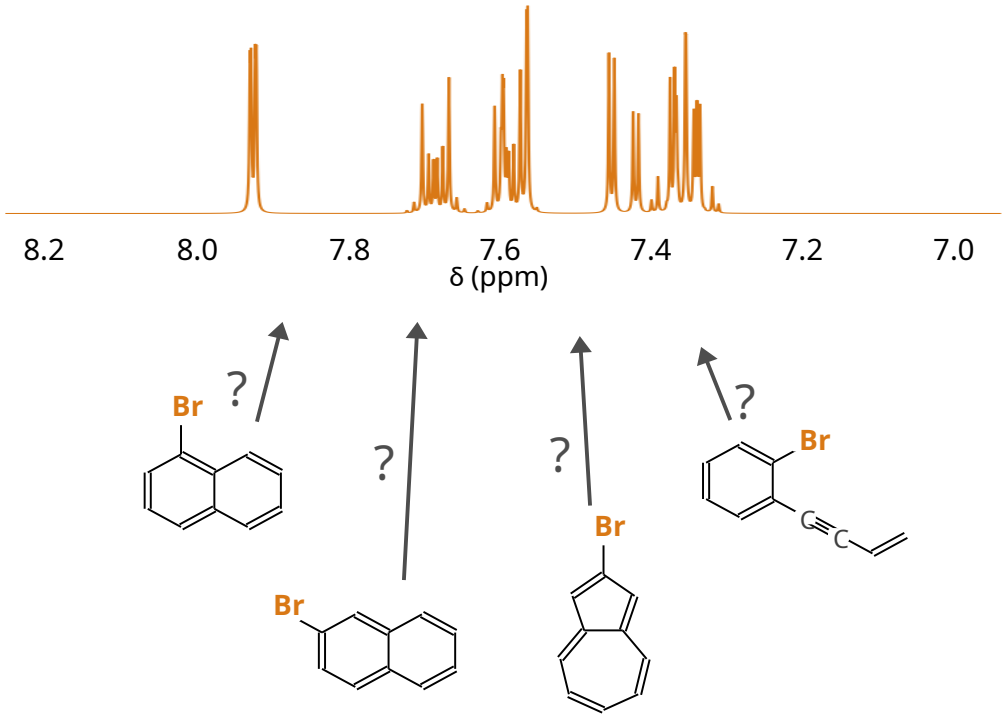




NMR spectroscopy



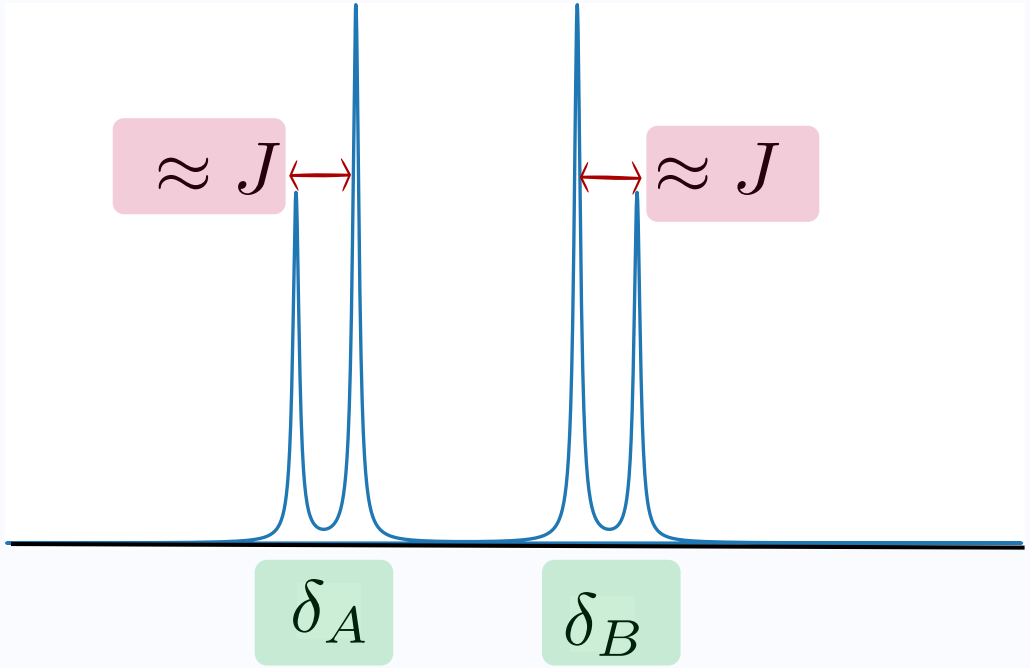
NMR spectrometer
(Image source: Wikipedia)



Simulating spectra for molecules, to compare to experimental results.

NMR Hamiltonian

Calcuate
NMR Hamiltonian



Magnetic field

spin-coupling

$$\hat{H} = -\sum_{\ell} \gamma_\ell B^z (1 + \delta_\ell)\, \hat{I}_\ell^z \;+\; 2\pi \sum_{k<\ell} J_{k\ell}\, \hat{\mathbf{I}}_k \cdot \hat{\mathbf{I}}_\ell$$

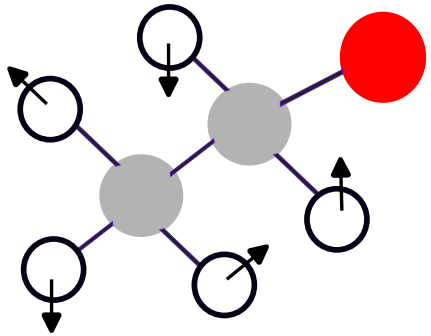
Gyromagnetic ratios

chemical shift

Pulse acquire experiment

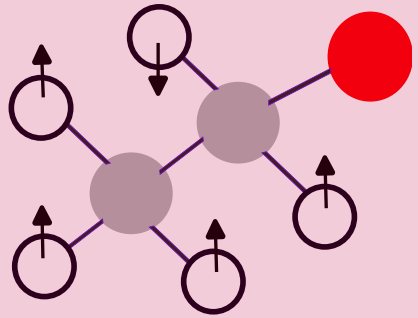
the before:

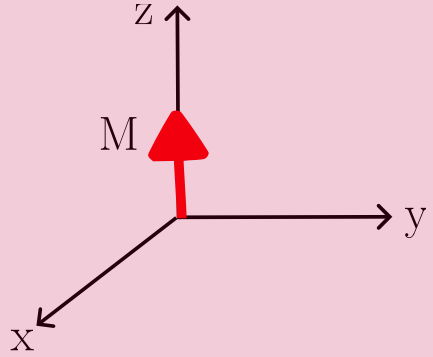
No net magnetization



1. Strong Magnetic field:

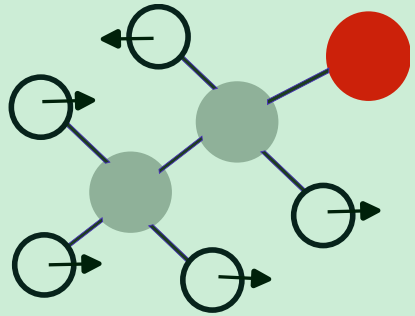
Magnetization emerges

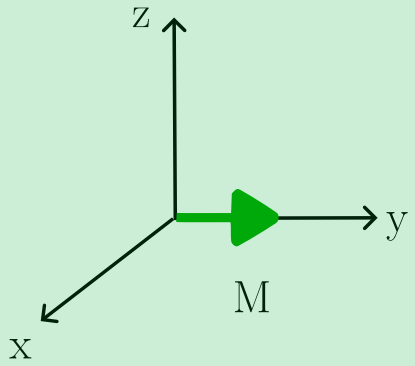




2. π/2 -pulse:

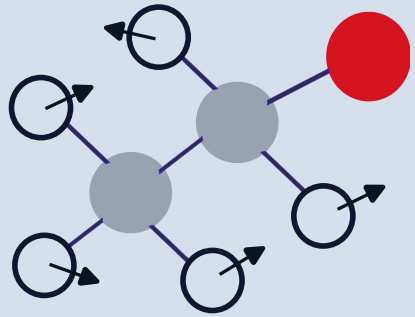
Magnetization flipps

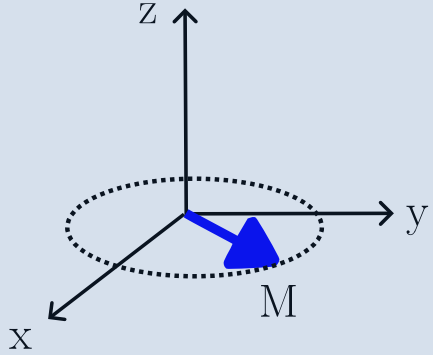




3. Acquisition

precession





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NMR spectroscopy as a use-case for QC

Measurement of 1D NMR spectrum

- Measured signal

$$\langle \gamma_k \hat{I}_k^x \rangle(t) = \mathrm{Tr}\left(\gamma_k \hat{I}_k^x \hat{\rho}(t)\right)$$

- Density-matrix at t_0

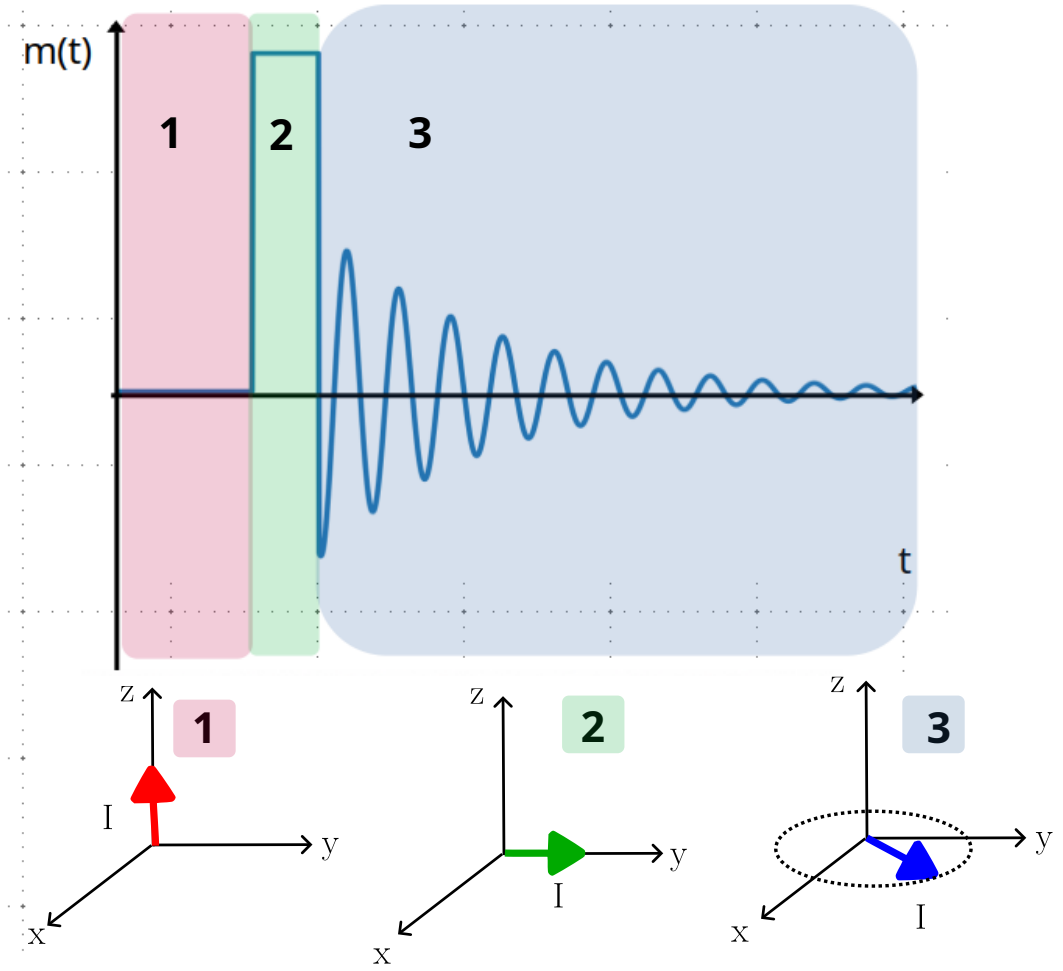
$$\hat{\rho}(t_0) = \frac{e^{-\beta\hbar\hat{H}}}{Z} \approx \frac{1}{Z} - \frac{\beta\hbar B^z}{Z} \sum_l \gamma_l (1+\delta_l) \hat{I}_l^z$$

- Time-evolution of density matrix

$$\hat{\rho}(t) \;=\; e^{-i\hat{H}t}\, \hat{\rho}(t_0)\, e^{\,i\hat{H}t}$$

- Hard pulse

$$\hat{\rho}(t) = e^{-i\hat{H}t} \hat{P}^x_{\pi/2} \hat{\rho}(t_0) (\hat{P}^x_{\pi/2})^\dagger e^{i\hat{H}t} \qquad \hat{P}^x_{\pi/2} = e^{-i\frac{\pi}{2}\hat{I}^x}$$



Documentation on HQStage: https://docs.cloud.quantumsimulations.de/hqs-spectrum-tools/index.html

NMR on a quantum computer

$$\langle \gamma \hat{I}^x \rangle(t) = \frac{\beta \hbar B^z}{Z} \gamma \sum_l \gamma_l (1+\delta_l) \mathrm{Tr}\left(\hat{I}^x e^{-i\hat{H}t} \hat{I}_l^y e^{i\hat{H}t} \right)$$

$$C_{yx}(t) = \mathrm{Tr}\left(\hat{I}^y(t)\, \hat{I}^x(0) \right)$$

Evaluate expectation value

$$\mathrm{Tr}\left(\hat{I}^y(t)\, \hat{I}^x(0) \right) = \sum_{n=1}^{2^N} \mathrm{Tr}\left(\hat{I}^y(t)\, \hat{I}^x\, |m_n^x\rangle\langle m_n^x| \right)$$

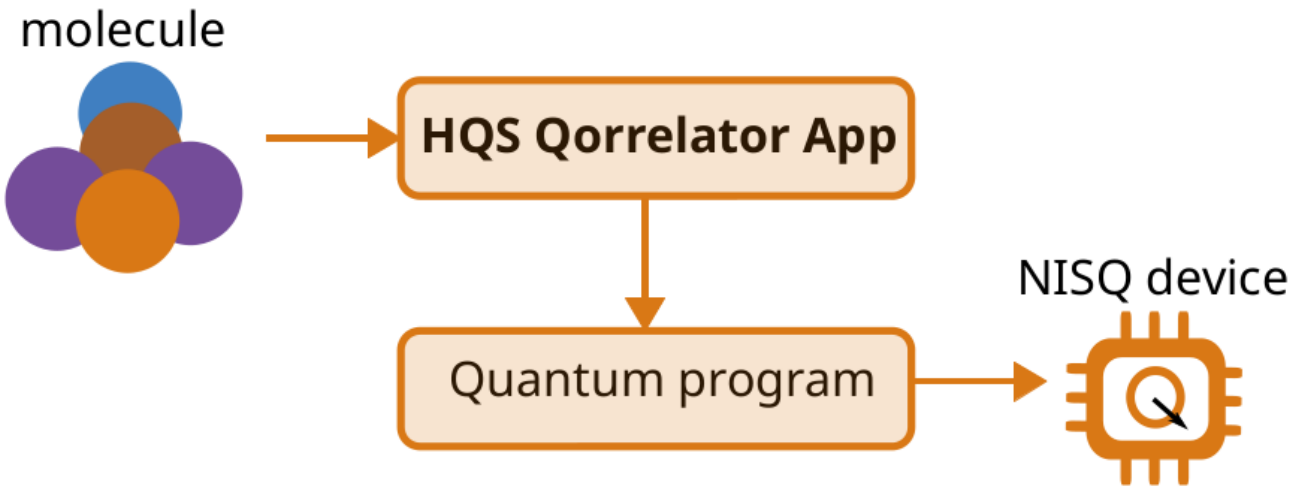
$$= \sum_{n=1}^{2^N} \mathrm{Tr}\left(\hat{I}^y(t)\, m_n^x\, |m_n^x\rangle\langle m_n^x| \right)$$

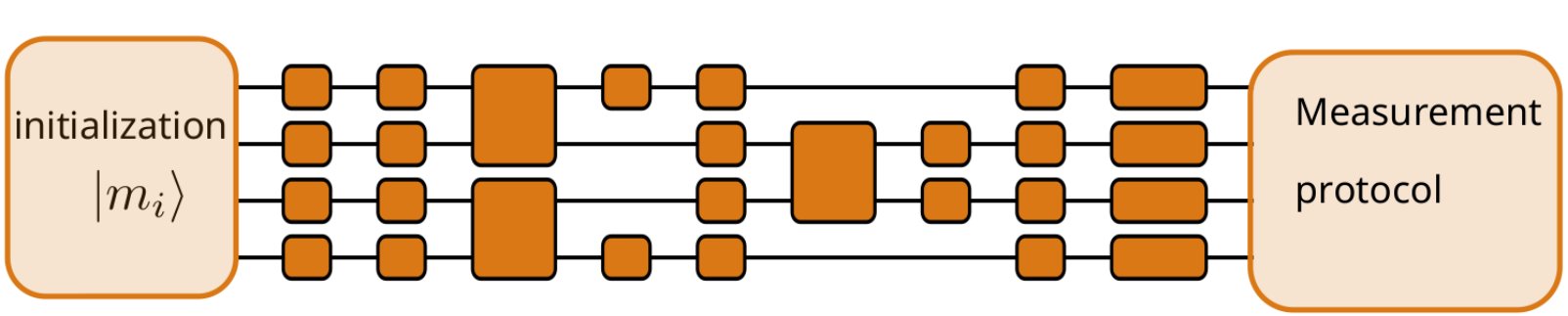
$$= \sum_{n=1}^{2^N} m_n^x\, \langle m_n^x | \hat{I}^y(t) | m_n^x \rangle$$

sum over exponentially many initial states!

A. Khedri et al., The impact of noise on the simulation of NMR spectroscopy on NISQ devices, arXiv:2404.18903 (2024)

Simulation of NMR on a QC



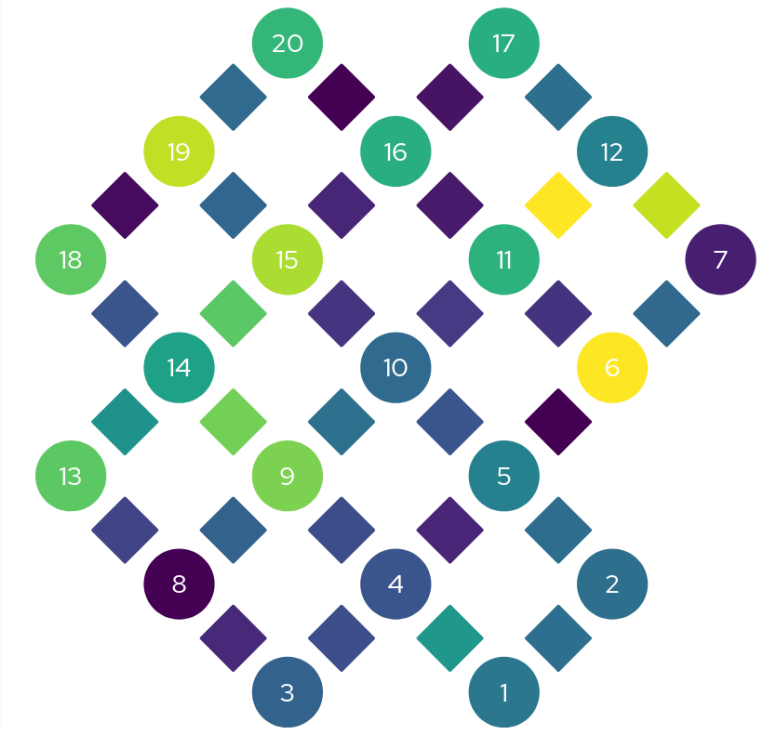


$\langle m_i(t)|S^x_{\mathrm{tot}}|m_i(t)\rangle$

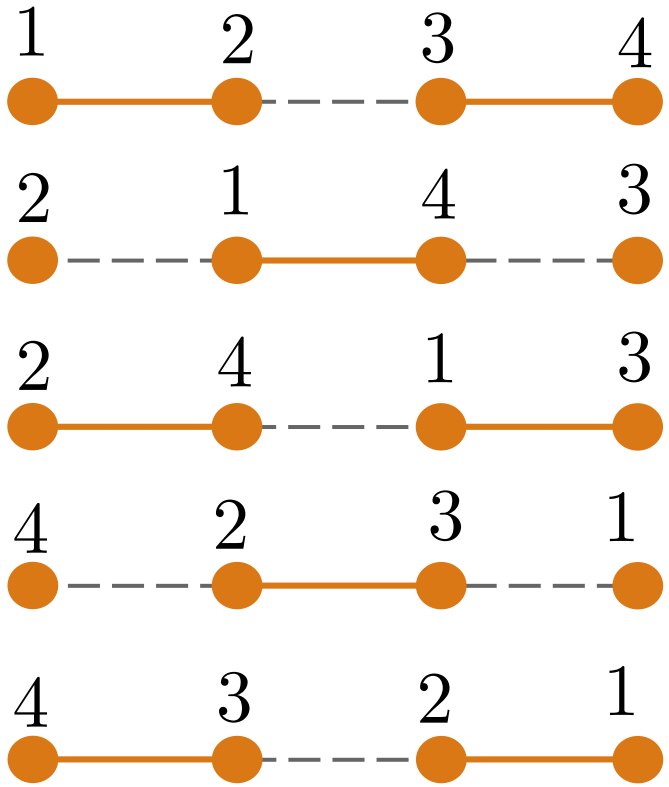
Repeat for all possible initializations and evaluate the sum

QSWAP algorithm

Nuclei interaction does not match connectivity of device



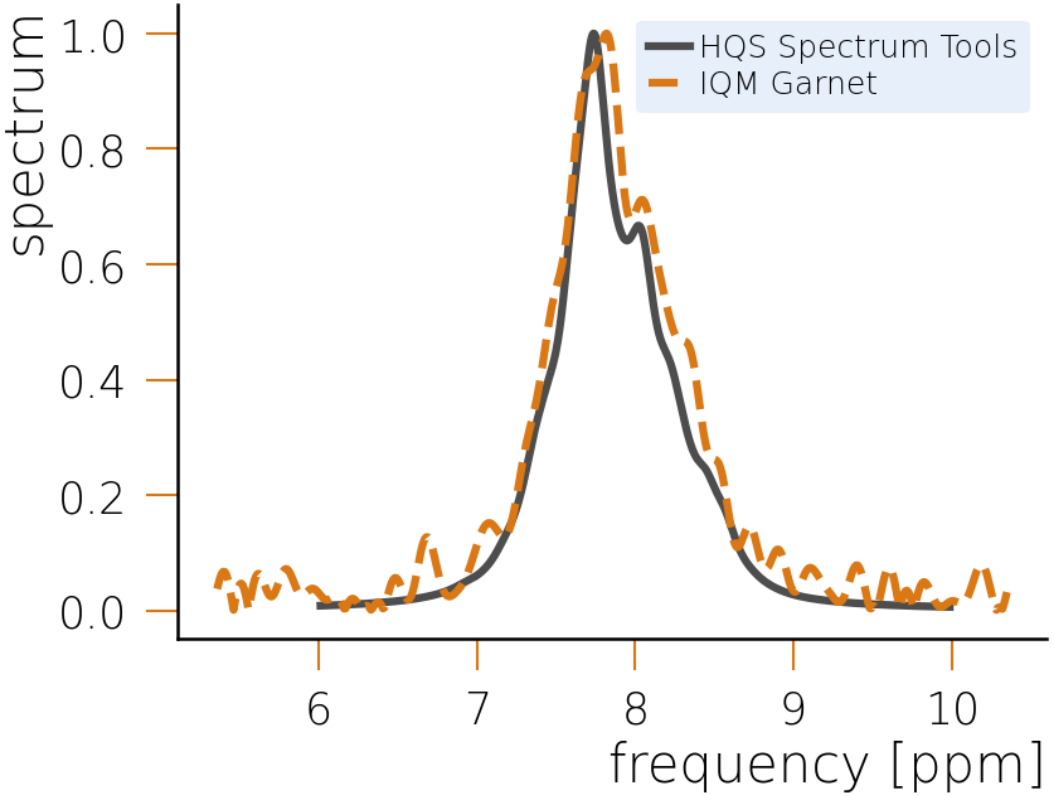
QSWAP: SpinInteraction + SWAP



After 1st Trotter order: Qubits are remapped

Ian D. Kivlichan et. al., Quantum Simulation of Electronic Structure with Linear Depth and Connectivity, Phys. Rev. Lett. 120, 110501 (2018)

Experiment on IQM Garnet



Molecule / NMR parameters

- C6H5NO2 (5 qubits)
- 20 MHz

Circuit / device parameters

- Trotter steps: 30
- 30 two-qubit gates per Trotter step
- Two-qubit depth: 15
- Two-qubit gate error: 0.005
- 1024 shots per time point
- Pauli twirling, readout error mitigation

What is a good use case?

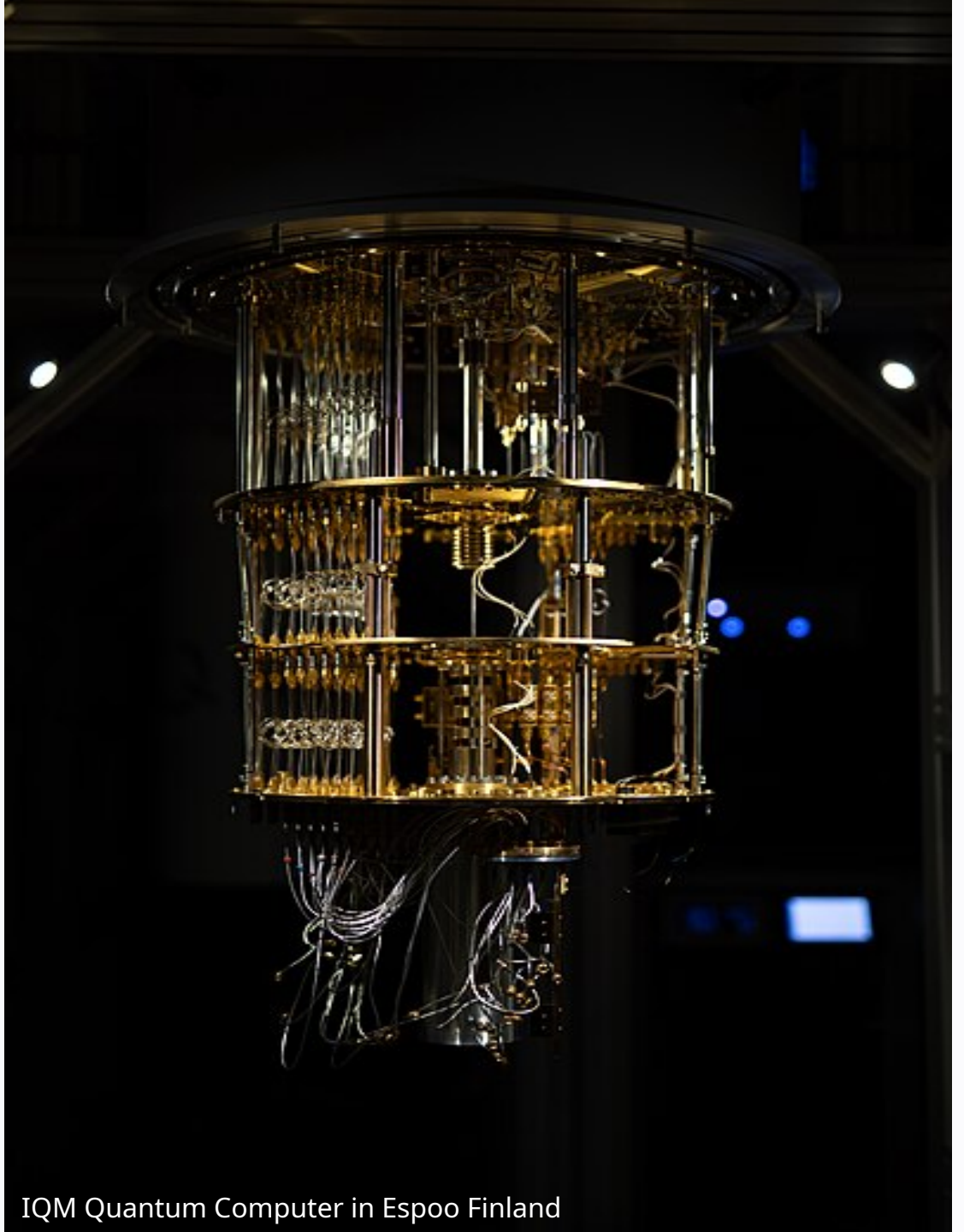
identify industry problem

transform to quantum

get the job done without the quantum computer

show quantum advantage

https://arxiv.org/abs/2506.15426



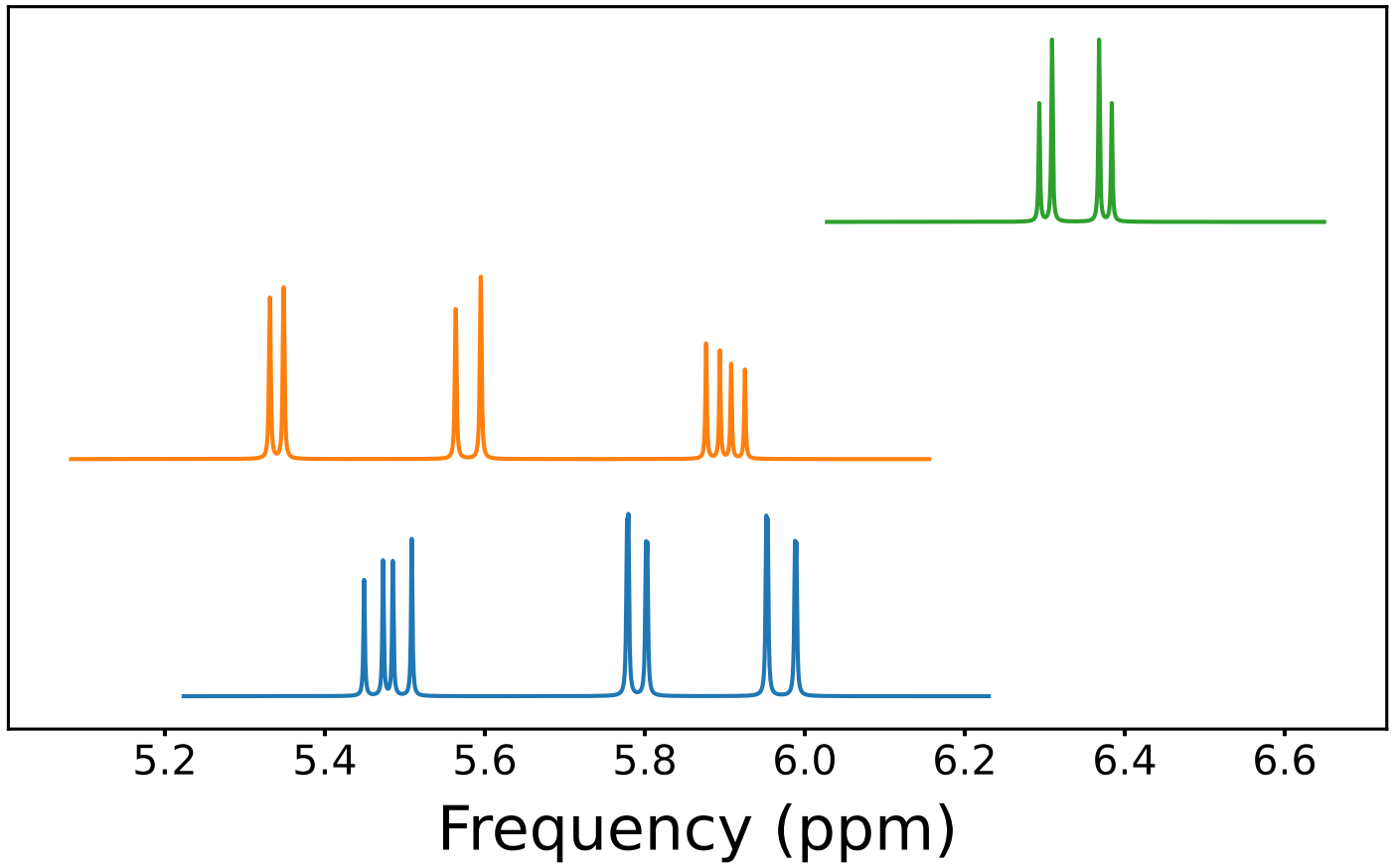
IQM Quantum Computer in Espoo Finland

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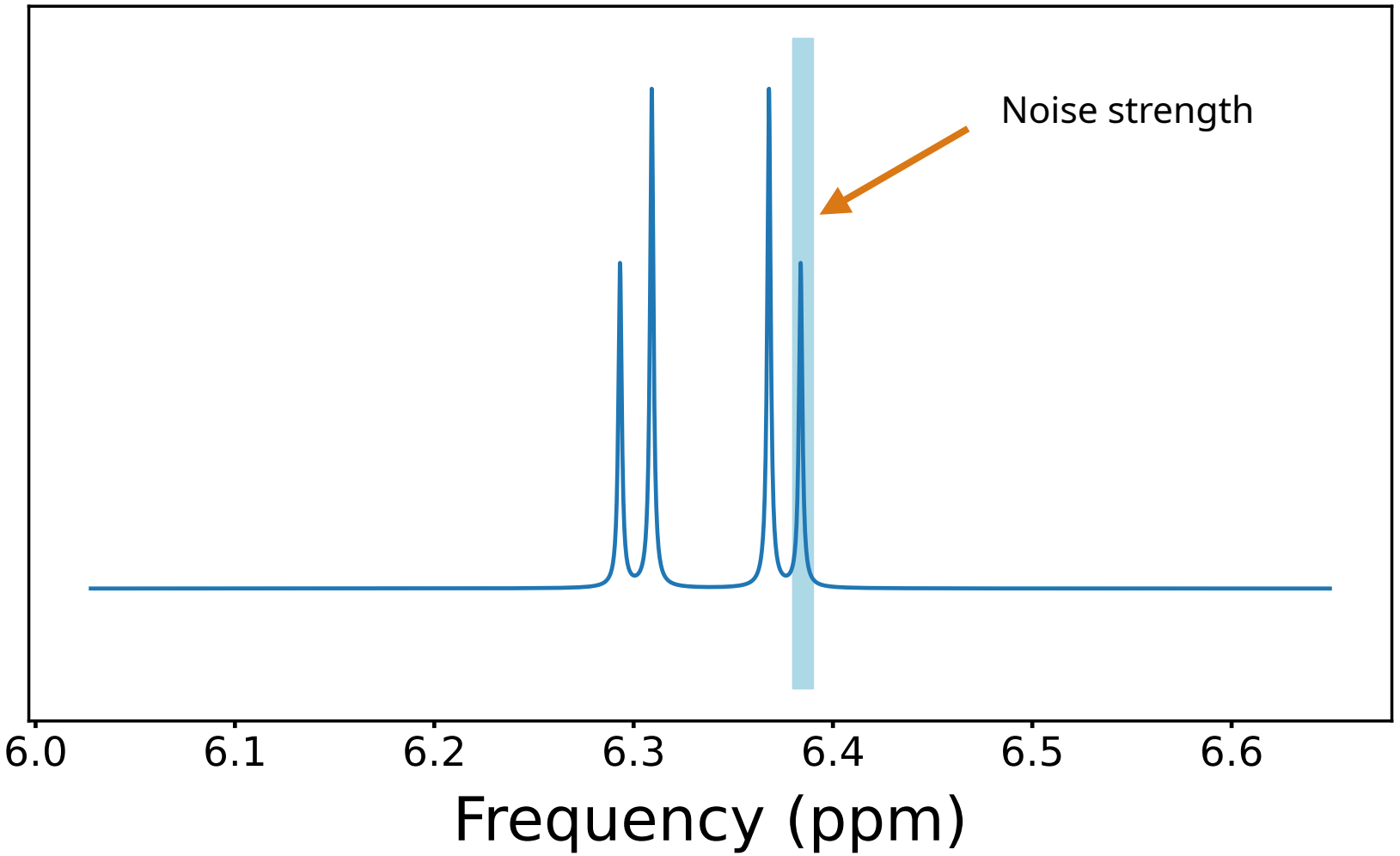
Hands-on Session

The Matching Task: Identifying Molecules with their NMR Spectrum

Three spectrum, three molecules, match them!



Explore the noise tolerence of NMR Spectroscopy





HQS QUANTUM SIMULATIONS

Thank you for your attention!

info@quantumsimulations.de

www.quantumsimulations.de